

Assignment 9, Social Science Inquiry II (SOSC13200-W26-3)

Friday 3/6/26 at 11:59pm

Collaboration: you may discuss conceptual questions with classmates, but your code and write-up must be your own.

Setup

Use the same data source you used in Assignment 8.

Load data

```
data_path <- "https://raw.githubusercontent.com/UChicago-pol-methods/climate-framing-replication/refs/h  
dat <- read.csv(data_path, as.is = TRUE)
```

Treatment labels (for convenience)

In this dataset: - `treatment_value = 0` is the control (no framing) - `treatment_value = 1..5` are the five frames

```
treatment_labels <- c(  
  "Control (no framing)",  
  "Negative science",  
  "Religious",  
  "Equity",  
  "Efficiency",  
  "Secular"  
)
```

1. PAP-to-paper crosswalk (short reading check)

Open: - <https://github.com/UChicago-pol-methods/climate-framing-replication/blob/main/preregistration/Pre-analysis%20plan.pdf>

Answer briefly (1–3 sentences each).

(1a) Outcome + arms

In the paper/PAP, what is the main outcome and how is it constructed? What are the treatment arms and what is the control condition?

```
# your answer here
```

(1b) Main estimands + “best arm”

What are the main causal quantities the PAP targets? In 2–3 sentences, explain what the “best arm” / policy-learning goal is and why sample splitting matters.

```
# your answer here
```

2. Measurement: rebuild the index outcome

The dataset includes `post_test` and `pre_test`. The PAP describes these as indices. In this section, you will rebuild them from the component items and confirm they match.

(2a) Recreate the post-test index

Create a new variable `post_index` that is the average of: - `gastax_after`, `carbtax_after`, `treaty_after`, `regcarb_after`

Then report the maximum absolute difference `max(abs(post_index - post_test))`.

```
# your code here
```

(2b) Recreate the pre-test index

Create `pre_index` as the average of: - `gastax`, `carbtax`, `treaty`, `regcarb`

Then report `max(abs(pre_index - pre_test))`.

```
# your code here
```

3. Multi-arm treatment effects (frame vs control)

In Assignment 8 you chose ONE contrast. Here you will estimate effects for **all frames vs control**. Use: - outcome: `post_test` - covariate: `pre_test` - treatment assignment: `treatment_value` (0 = control; 1–5 are frames)

For this assignment, **do not** do additional cleaning / subsetting beyond dropping missing values in the variables you use.

(3a) Create a results table

For each arm `k` in {1,2,3,4,5}, compare arm `k` to control (0) and compute:

- 1) Unadjusted estimate: difference in means (arm `k` minus control)
- 2) Adjusted estimate: regression `post_test ~ W + pre_test` on the arm-`k`-vs-control subset

Build a data frame called `results` with one row per arm and columns: - `arm` (a label like “Scientific”, “Religious”, etc.) - `ate_unadj` - `ate_adj` - `se_adj` - `ci_low_adj`, `ci_high_adj` (use normal approx: estimate $\pm 1.96 * SE$)

Print `results`.

```
# your code here
```

(3b) One-sentence interpretation

In one sentence: which arm looks best by the adjusted estimate, and how large is the effect (in index units)?

```
# your answer here
```

4. “Best arm” estimation with simple sample splitting (selective inference)

If you pick “the best” arm using the full sample and then report its estimated effect using the same sample, that estimate is (on average) too optimistic because you selected it based on noise.

Here you'll implement a simplified sample-splitting procedure like the one described in the PAP/paper.

(4a) Choose the best arm on a training set

Randomly split the data into two halves: “train” and “test” (use `set.seed()`).

On the training half only, run the multi-arm regression:

$$post_test_i = \alpha + \sum_{k=1}^5 \tau_k \mathbf{1}\{treatment = k\} + \gamma pre_test_i + \varepsilon_i,$$

where treatment 0 (control) is the baseline category.

Identify which arm k has the largest estimated $\hat{\tau}_k$ on the training set. Save it as an integer called `best_k_train`.

```
# your code here
```

(4b) Estimate the best arm effect on the test set (held out)

Using the test half only: - keep only observations in control (0) and the chosen best arm `best_k_train` - create $W = 1$ for best arm, 0 for control - estimate the adjusted effect via `lm(post_test ~ W + pre_test)`

Report the estimated effect (coefficient on W) and its SE.

```
# your code here
```

(4c) Compare to the naive (non-split) estimate

Now do the naive version on the full sample: - pick the best arm using the full sample (same multi-arm regression) - estimate that arm's adjusted effect vs control using the full sample

Report (i) which arm is “best” under the naive approach and (ii) the estimated effect. In 2–3 sentences: compare the naive estimate to the split-sample estimate and explain why they may differ.

```
# your code here + your answer here
```